

Shadows: A Laboratory to probe disk thermodynamics and hydrodynamics

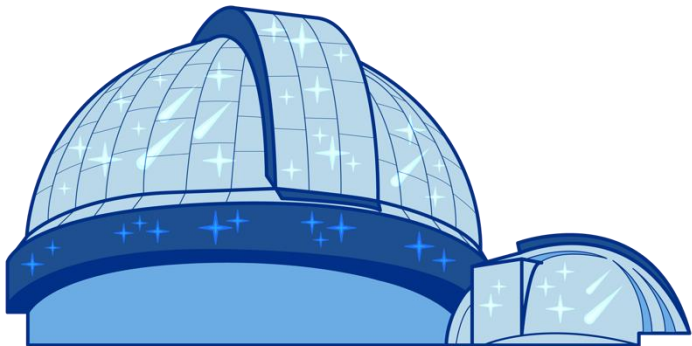
Shangjia Zhang

NHFP Sagan Fellow

Columbia University

June 9, 2026

3rd PFITS+ Meeting on Hydrodynamics of
Circumstellar Disks and Planet Formation



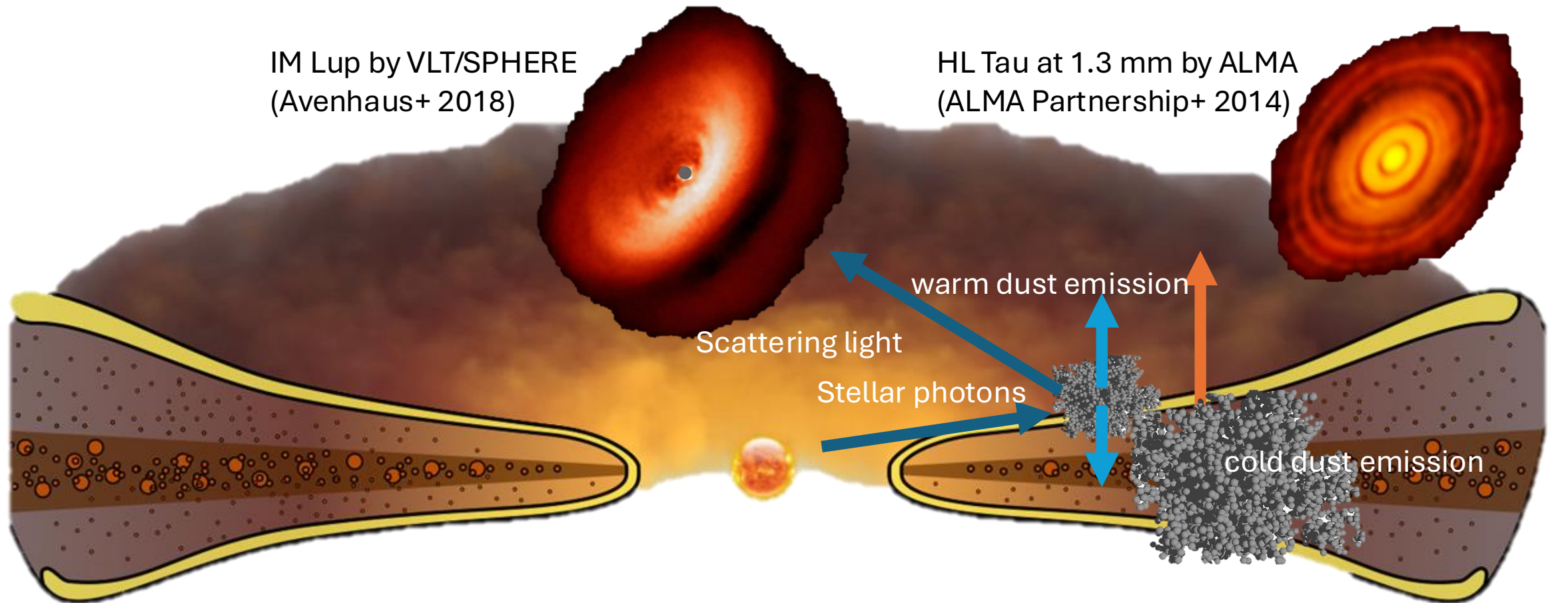
 **COLUMBIA UNIVERSITY**
ASTRONOMY AND ASTROPHYSICS



NASA Hubble
Fellowship Program

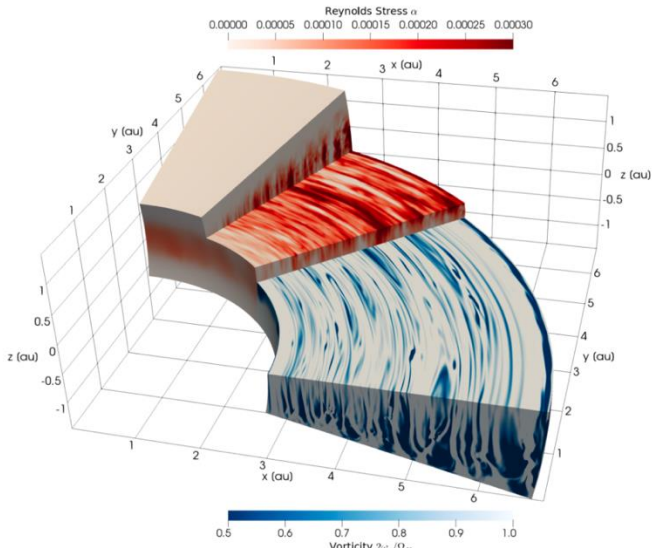
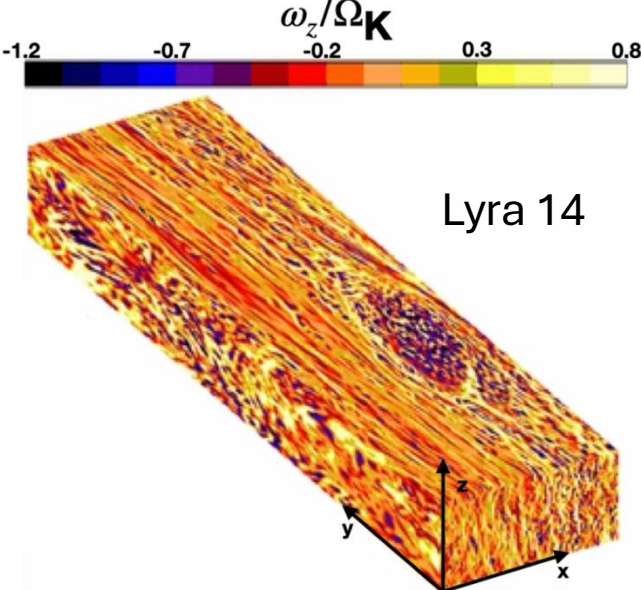
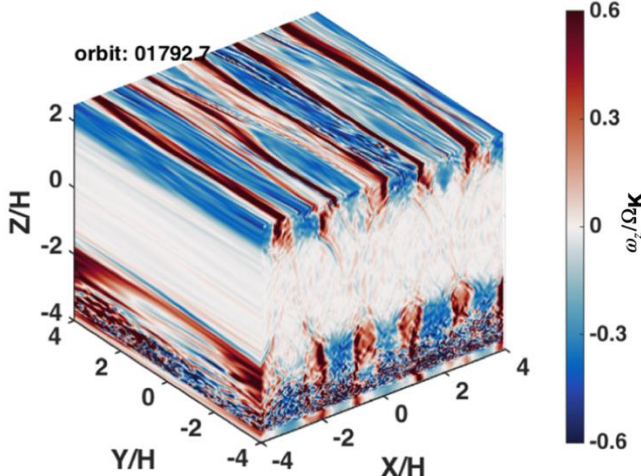
Radiative transfer

- In a passive disk
- Heating: stellar irradiation
- Cooling: dust radiative cooling



Thermodynamics

Also important in planet-disk interactions and GI

Vertical Shear Instability	Convective Overstability	Zombie Vortex Instability
 Reynolds Stress α Vorticity $2\omega_z/\Omega_K$	 Lyra 14 ω_z/Ω_K	 orbit: 01792.7 ω_z/Ω_K
$\tau_{\text{cool}}\Omega_K \ll 1$	$\tau_{\text{cool}}\Omega_K \sim 1$	$\tau_{\text{cool}}\Omega_K \gg 1$
$q \neq 0$	$-1 < p/q < 1/(\gamma - 1)$	$ z \gtrsim \sqrt{\gamma/(\gamma - 1)}H$
$\alpha_S \sim 10^{-4}$	$\alpha_S \sim 10^{-3}$	$\alpha_S \sim (10^{-5} - 10^{-4})^\dagger$
Outcome: turbulence & vortices		Lesur+ 23

This afternoon

Thomas Pfeil, Jip Matthijsse, Nadine Soliman, Yuya Fukuhara, Karim Shariff Lixandra Flores-Rivera on Monday, David Melon-Fuksman on Thursday

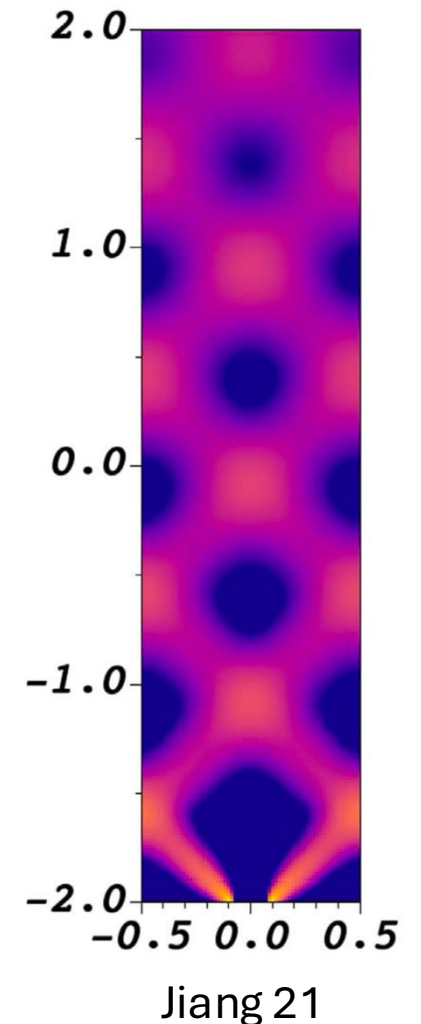
Marius Lehmann on Thursday Min-Kai Lin on Friday

Chen Wang on Wednesday

Radiation-Hydrodynamics

- How do we capture both the thermal structure and thermodynamics?
- Athena++ (Stone+ 20) and implicit radiation scheme (Jiang 21)
 - Solves the time-dependent radiative transfer equation for the specific intensity at every timestep (Jiang+ 14)
 - Multi-frequency (Jiang 22)
 - Works at both optically thin and thick regimes
 - Treats shadow and beam crossing properly

Stanley Barnoett talk on Friday



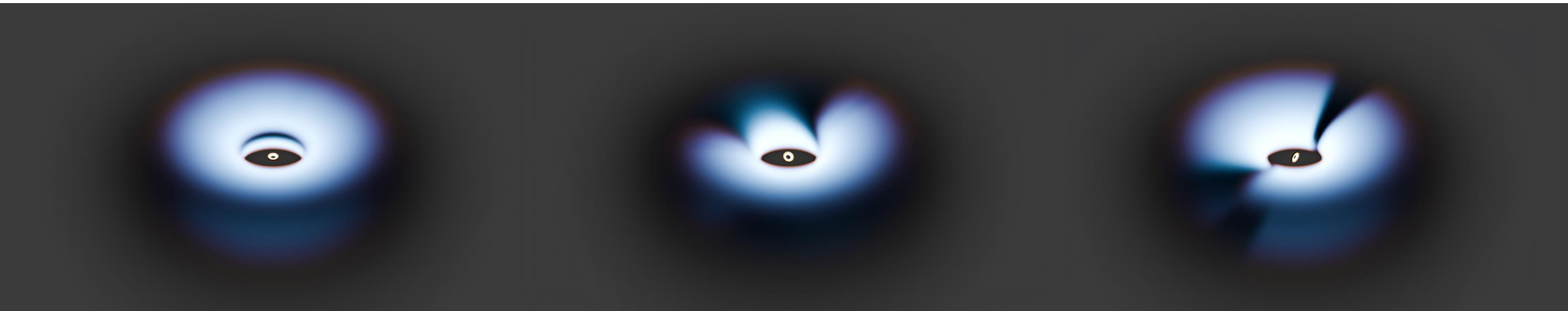
What if the irradiation is not axisymmetric?

- There are many observations evidences.

$i = 0^\circ$

$i = 30^\circ$

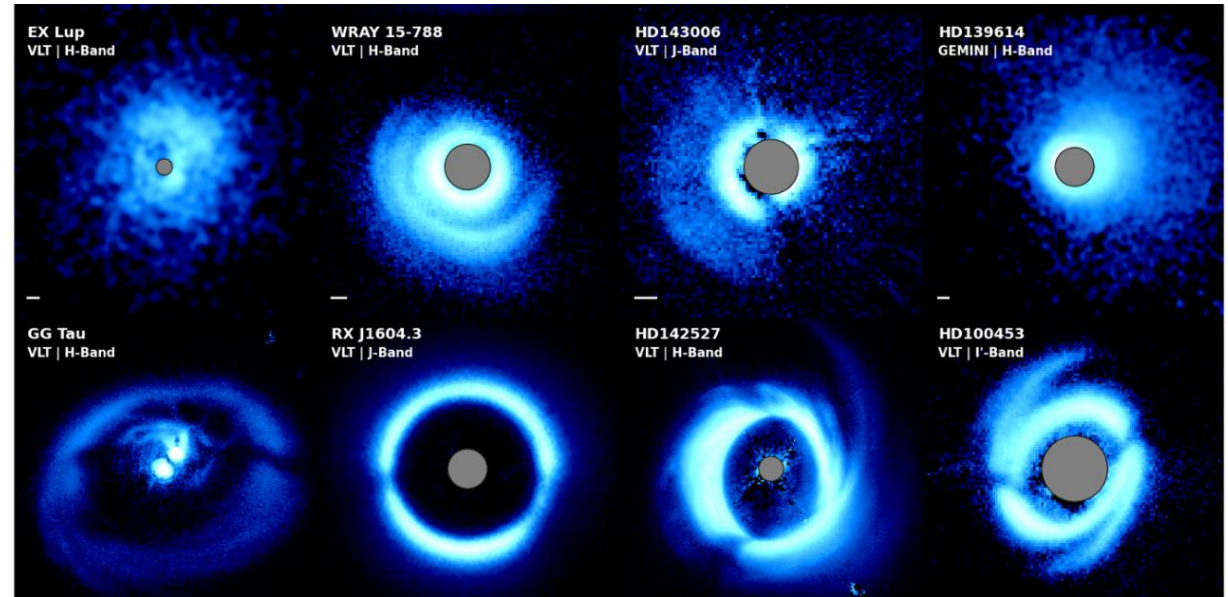
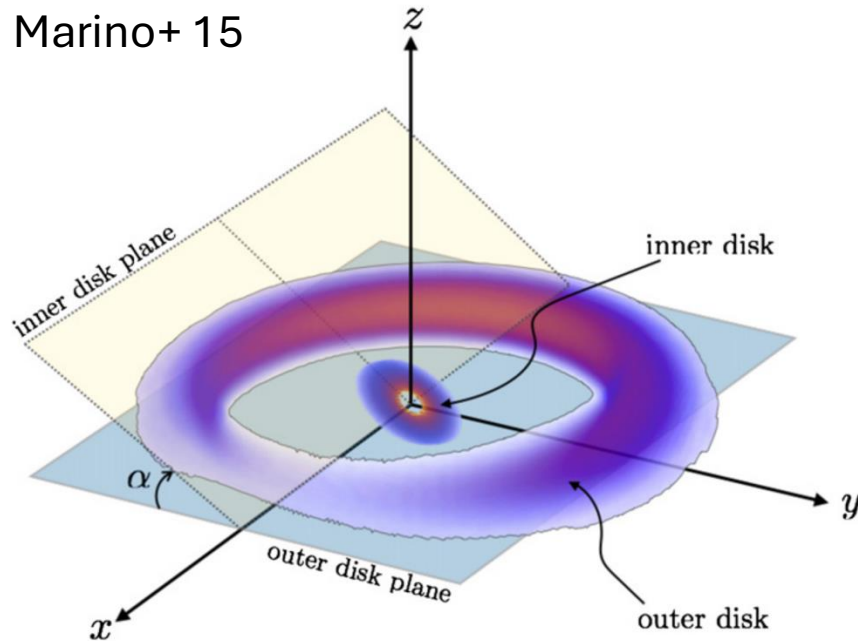
$i = 90^\circ$



Misalignments from NIR scattered light

- Shadow lanes cast by misaligned inner disk are common

Marino+ 15

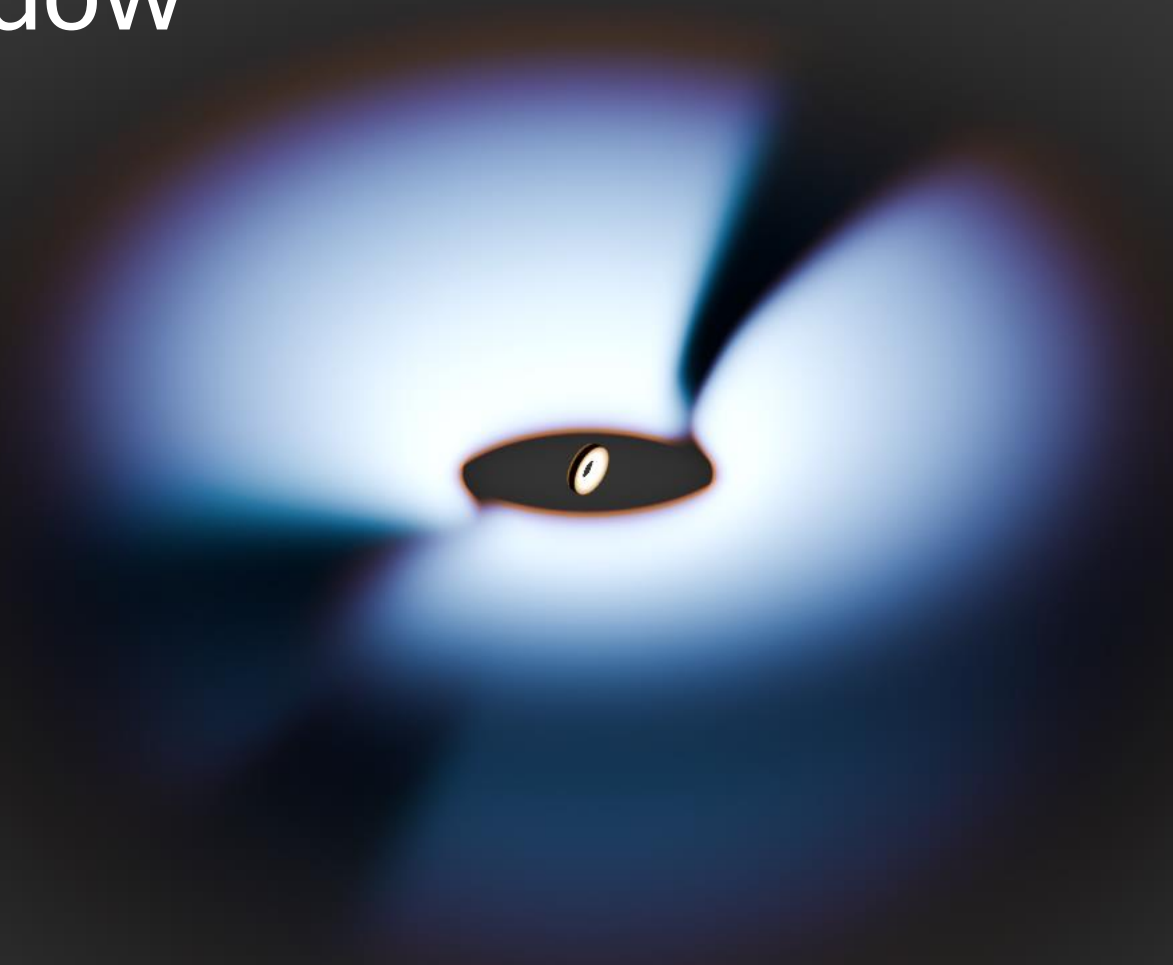


Benisty+ 23

Coplanar Shadow

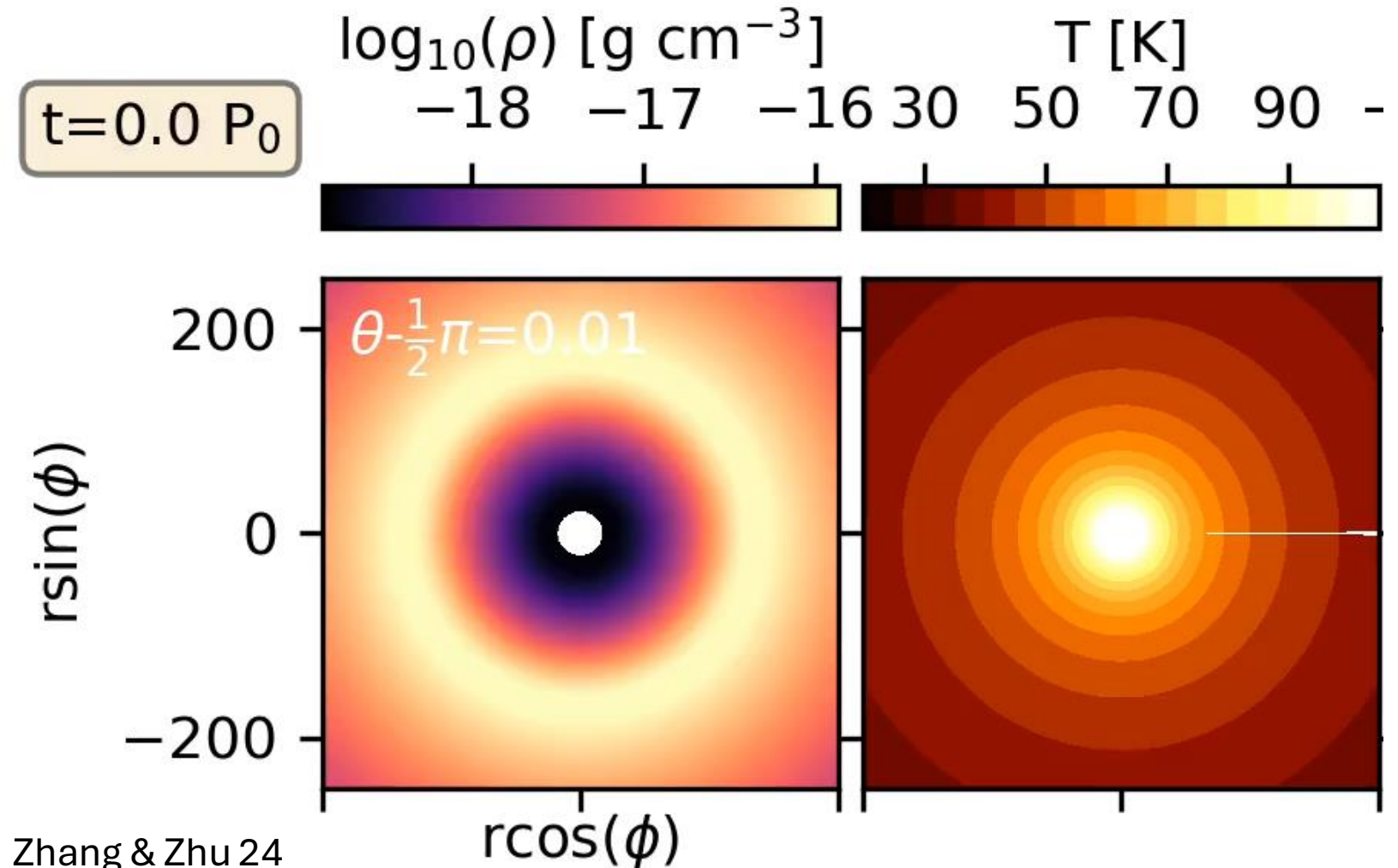


Polar Shadow



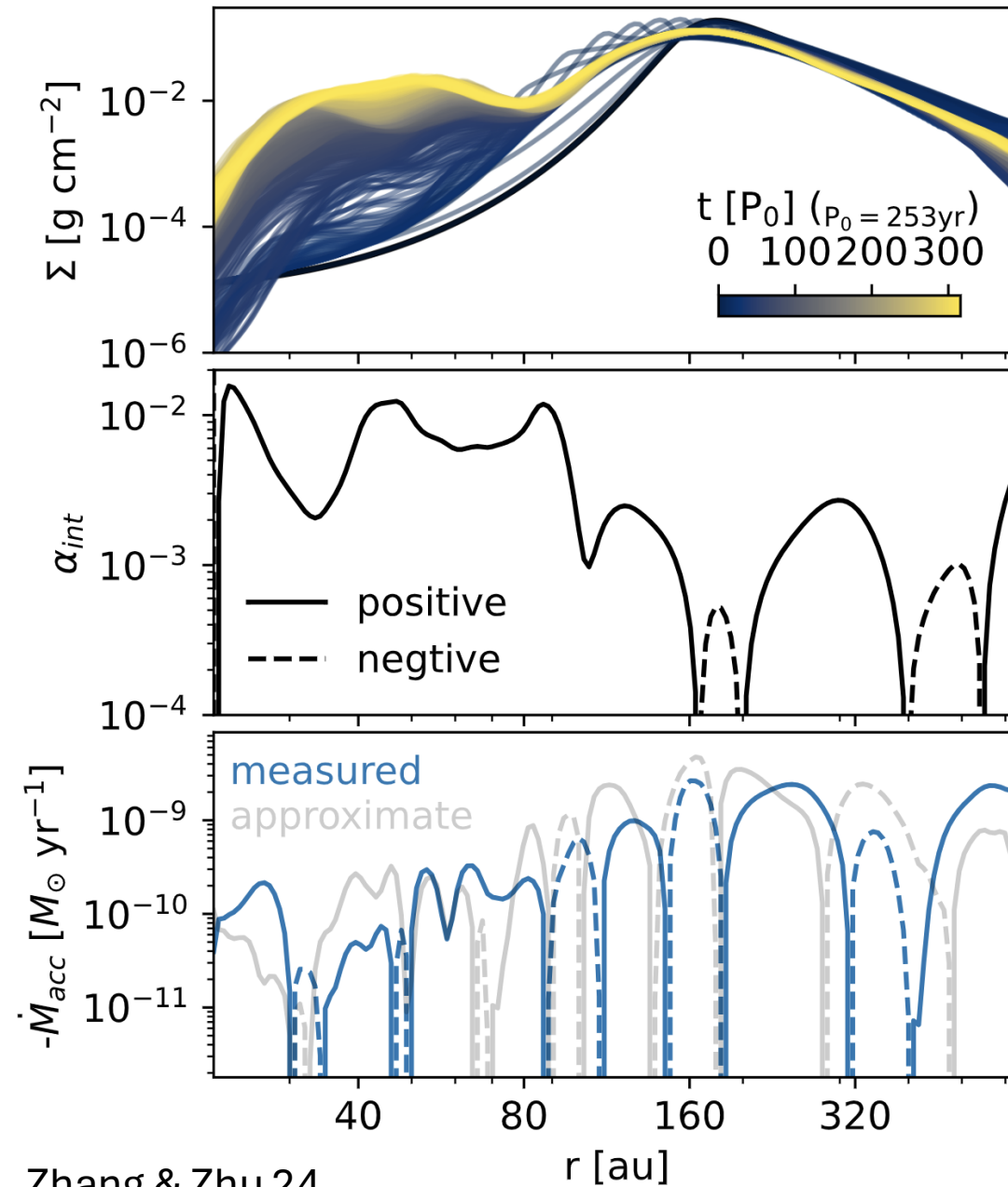
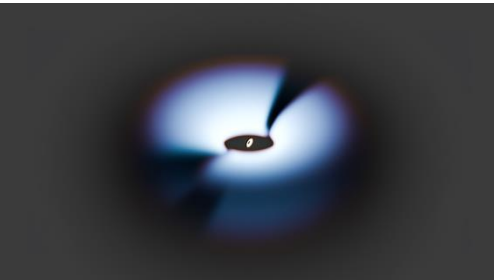
Polar Shadow

- 3D radiation-hydro simulations
- Inner disk is perpendicular to the outer disk
- Non-precessing



Polar Shadow

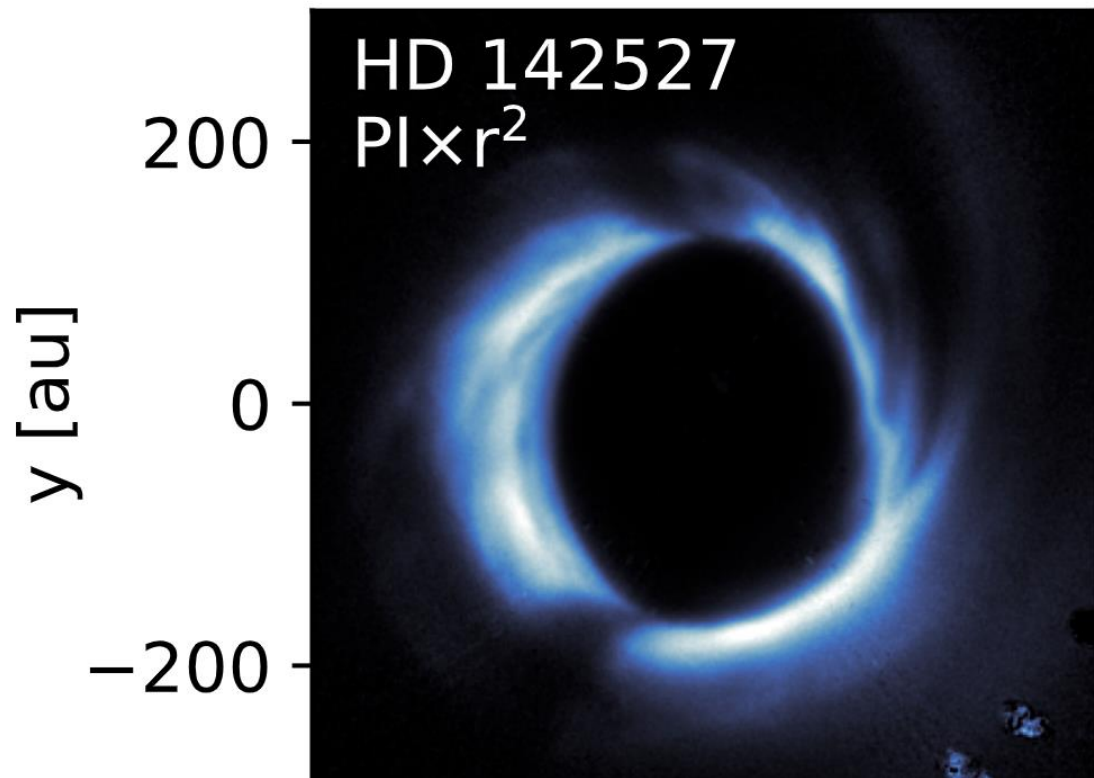
- Lead to accretion in the cavity
- $\alpha \sim 10^{-2}$ in the cavity
- $\alpha \sim 10^{-3}$ at the ring



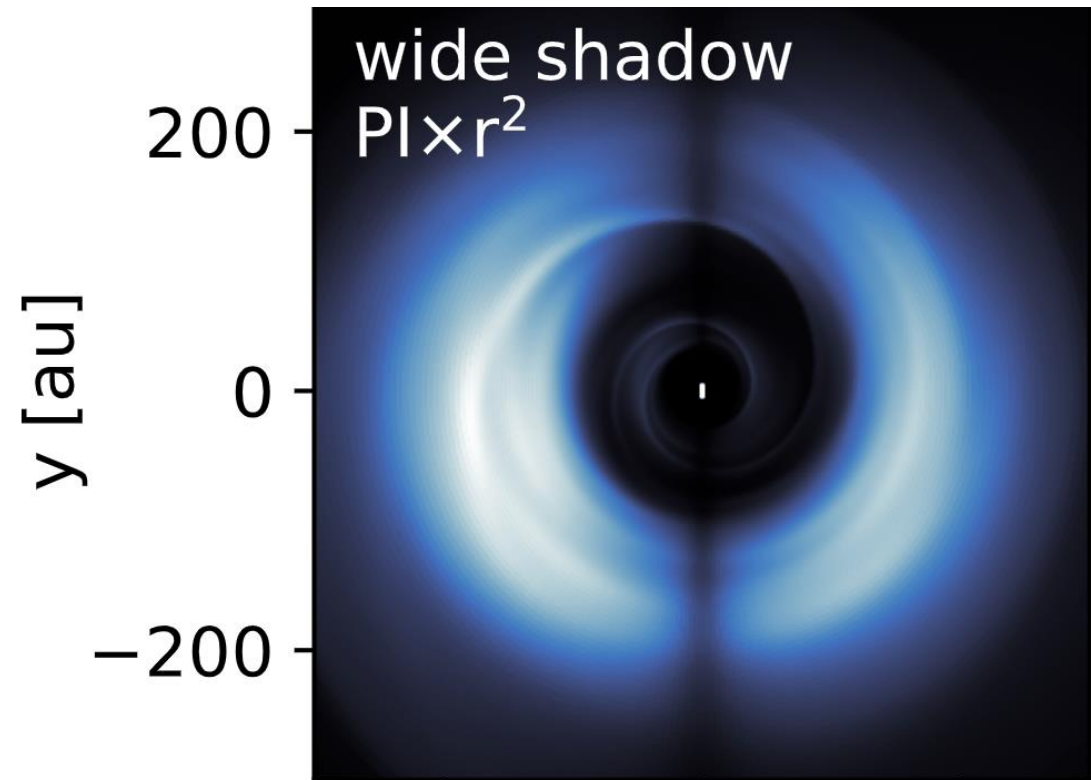
Zhang & Zhu 24

Polar Shadow

- Spirals can be launched by shadows

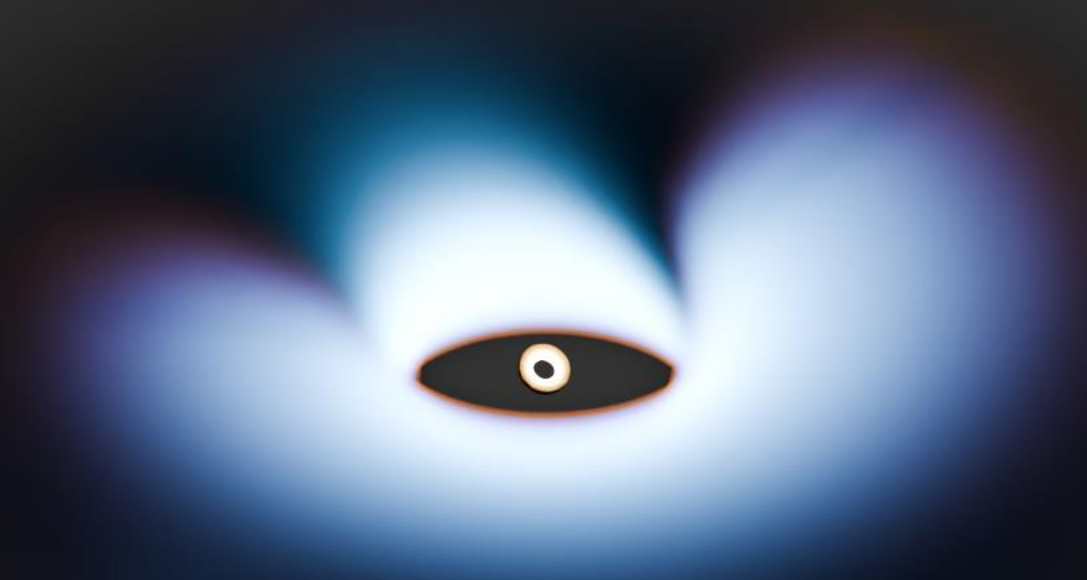


VLT/SPHERE (Hunziker+ 2021)



Rad-Hydro + RADMC-3D (Zhang & Zhu 24)

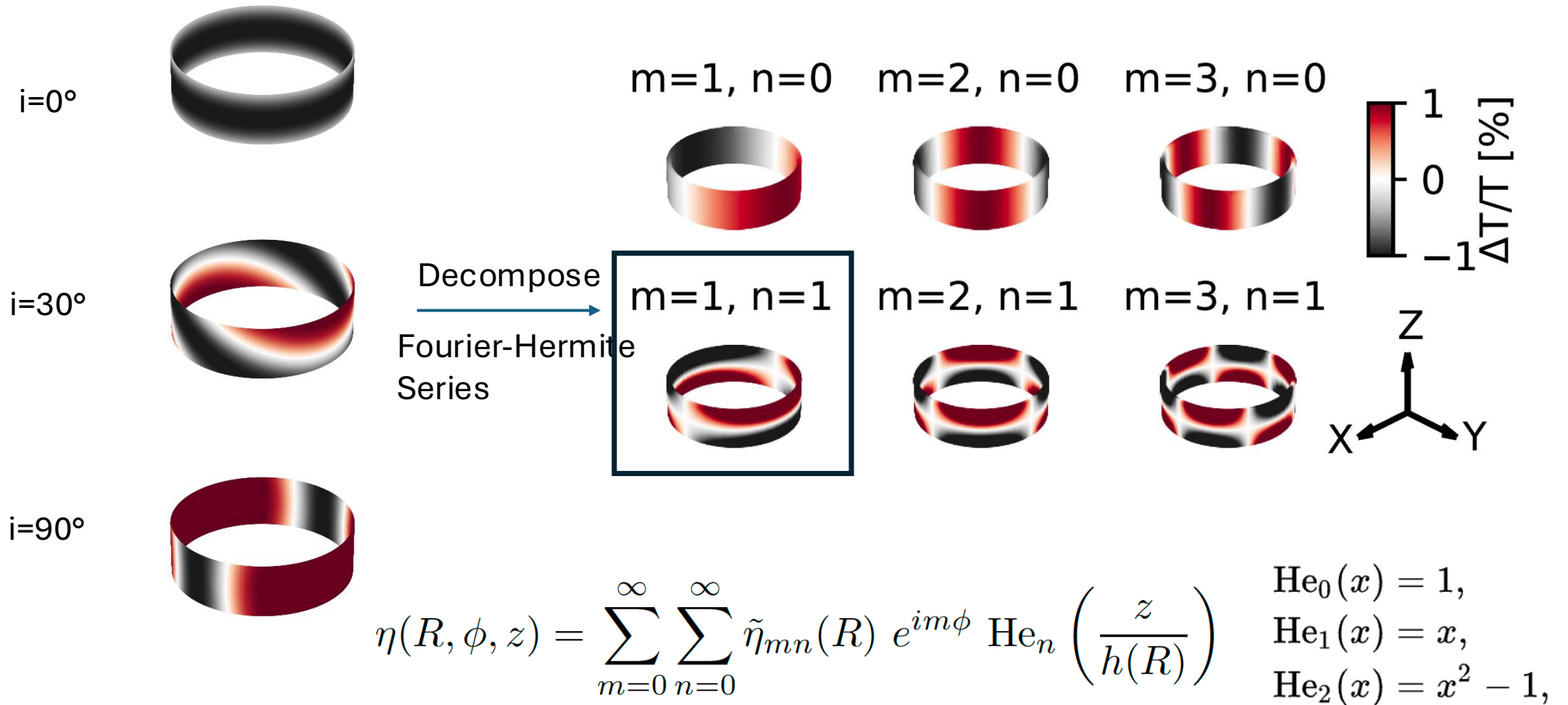
What above a mildly inclined disk?



Credit: Shangjia Zhang (Columbia University)

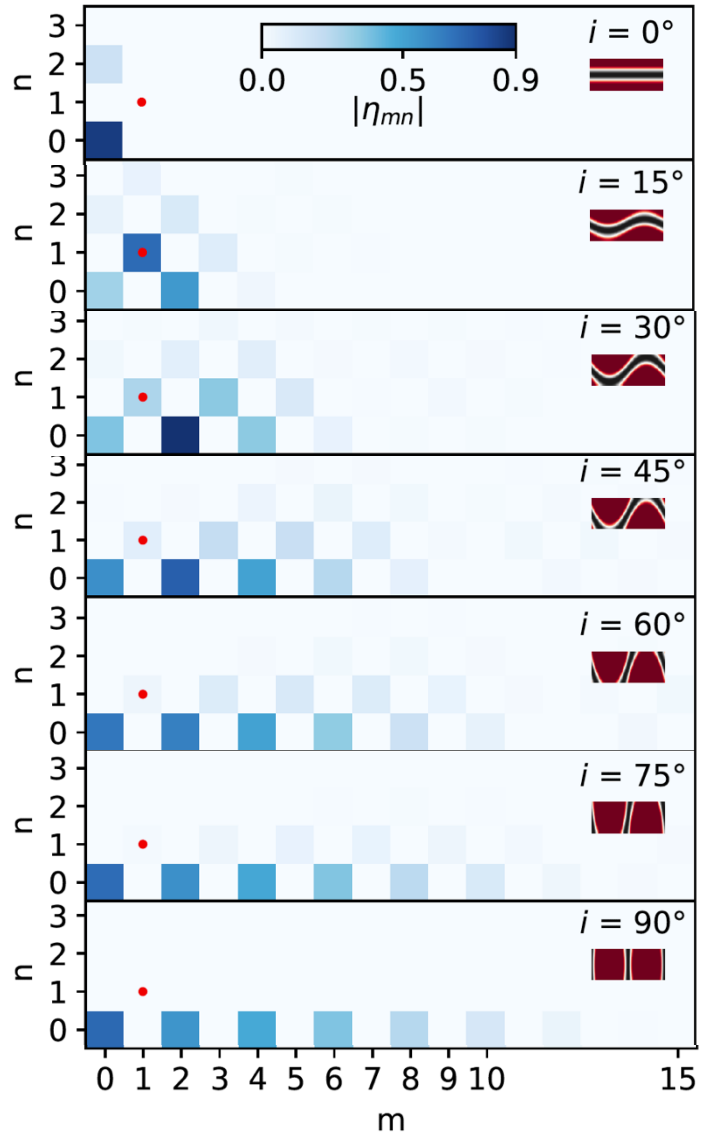


Why warps?

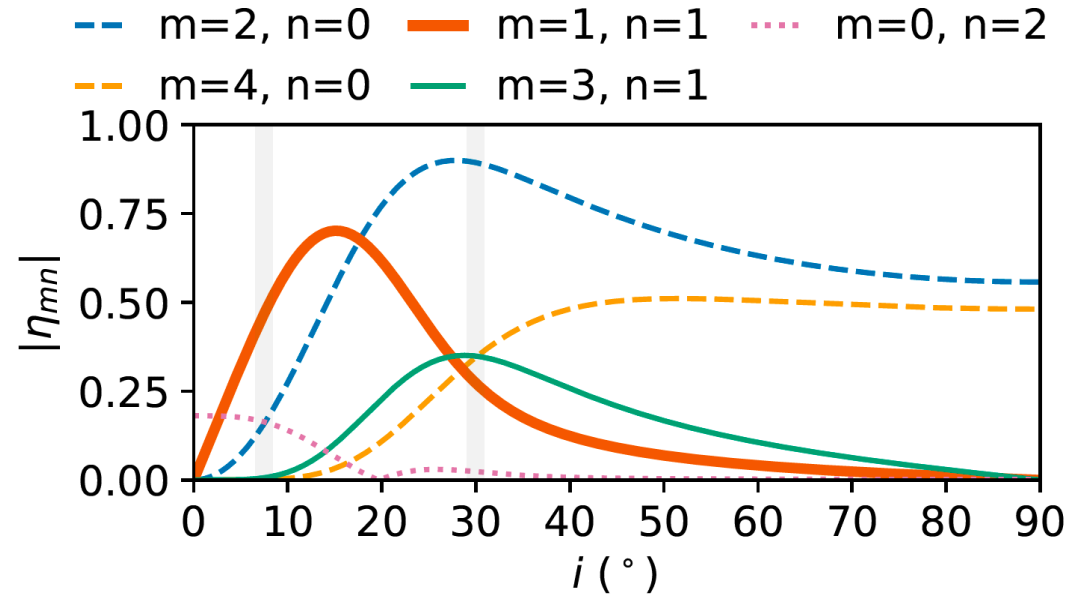


Fourier-Hermite decomposition

$$\eta(R, \phi, z) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \tilde{\eta}_{mn}(R) e^{im\phi} \text{He}_n \left(\frac{z}{h(R)} \right)$$



- The $m=1, n=1$ amplitude is significant between $i=3-30^\circ$.



$m=1, n=1$ temperature perturbation warps the disk

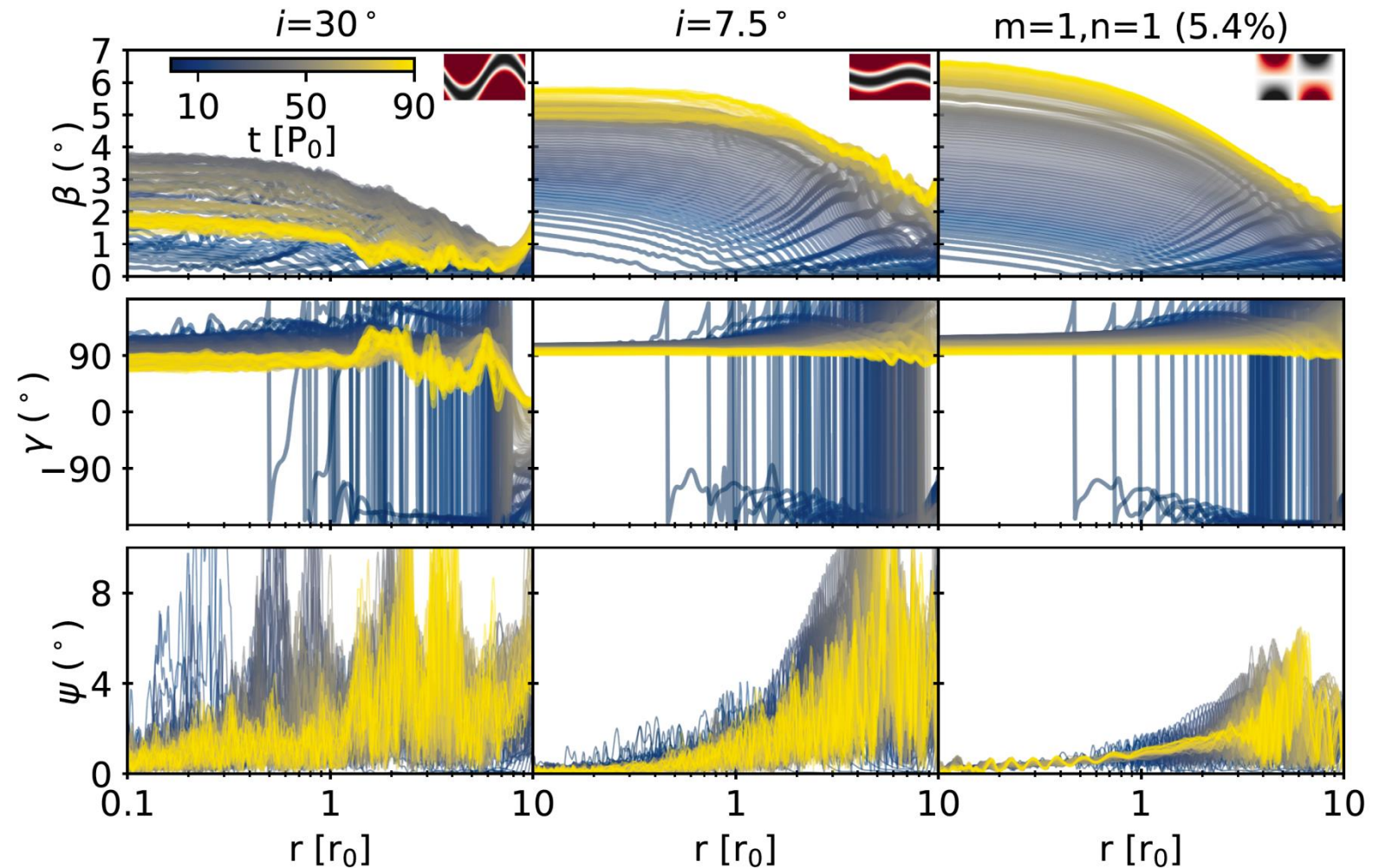


$$T(r, \theta, \phi, t) = T_{\text{bg}}(R) \times f(\theta_s, t).$$

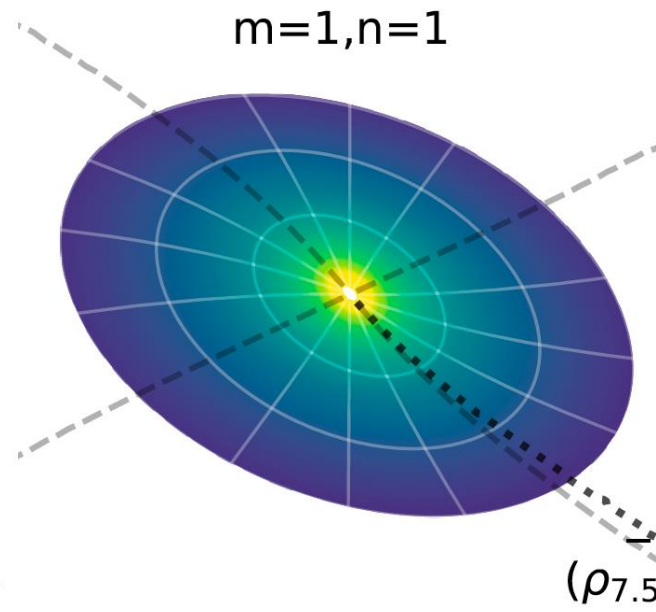
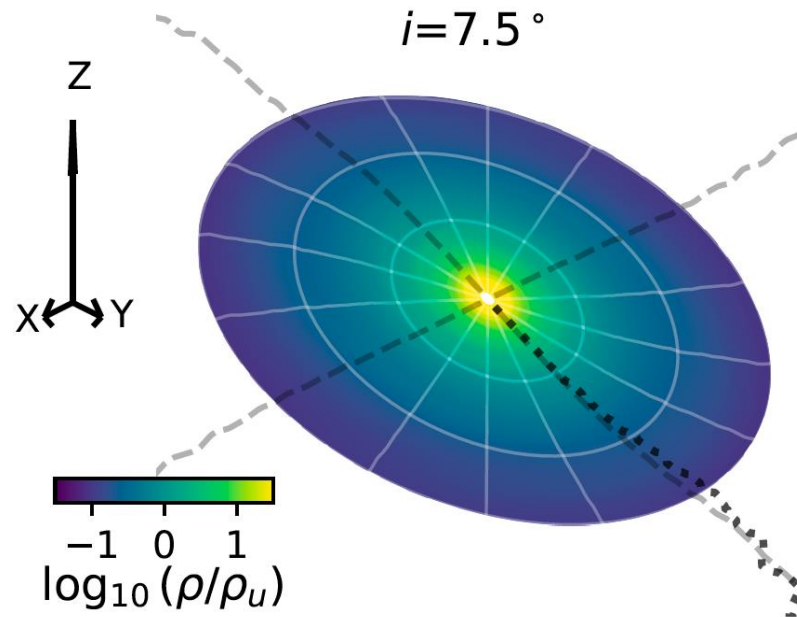
$$f(\theta_s, t) = 1 + A(t) g(\theta_s),$$

$$g_{11}(R, \phi, z) = \frac{z}{h(R)} \sin \phi,$$

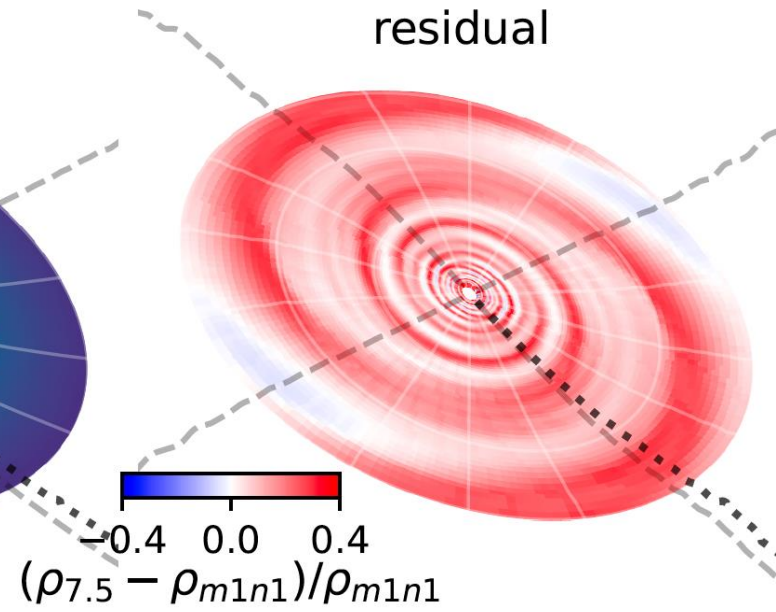
$$\Sigma_g = \Sigma_{g,0} \left(\frac{R}{R_0} \right)^{-1}$$



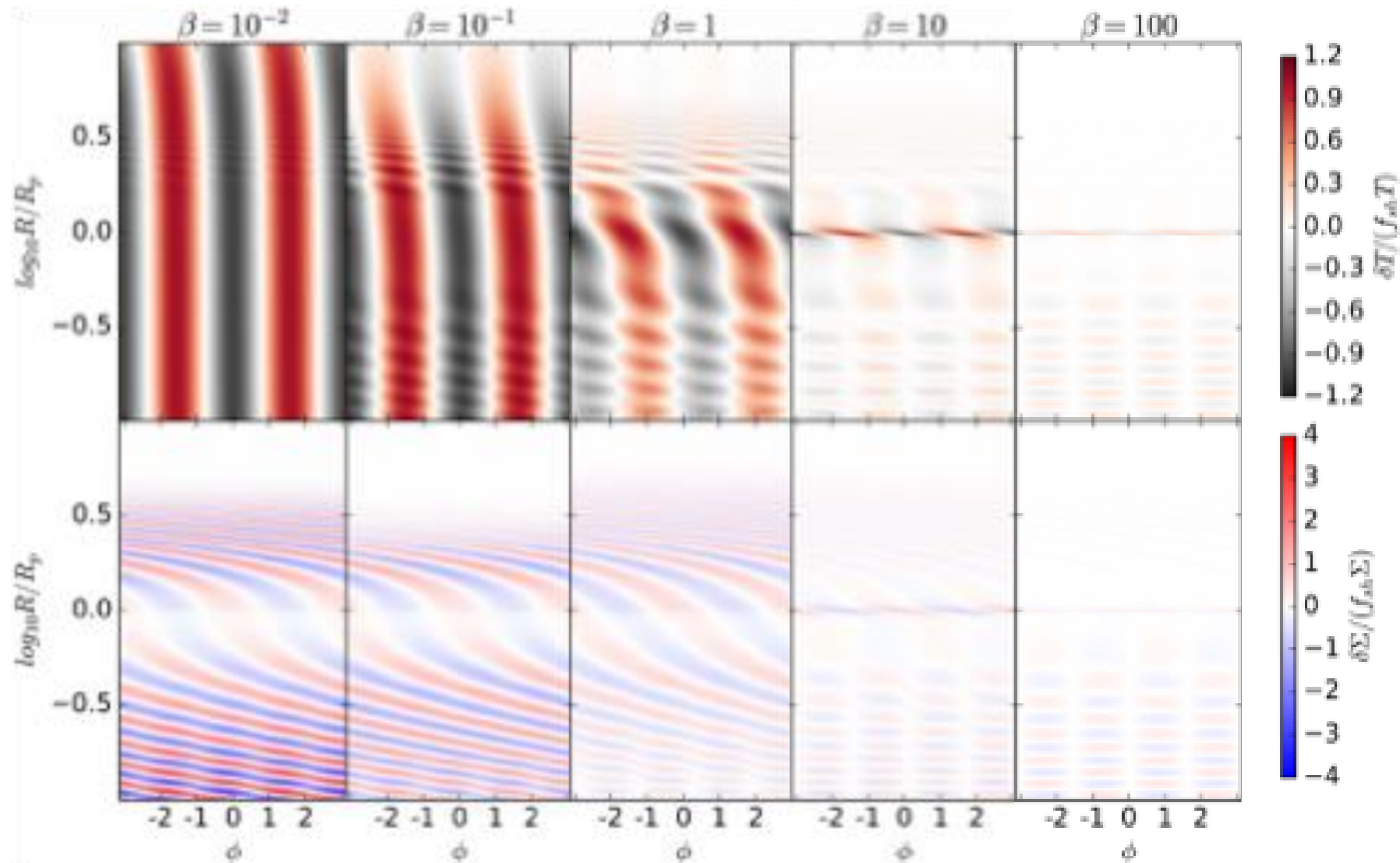
$m=1, n=1$ temperature perturbation warps the disk

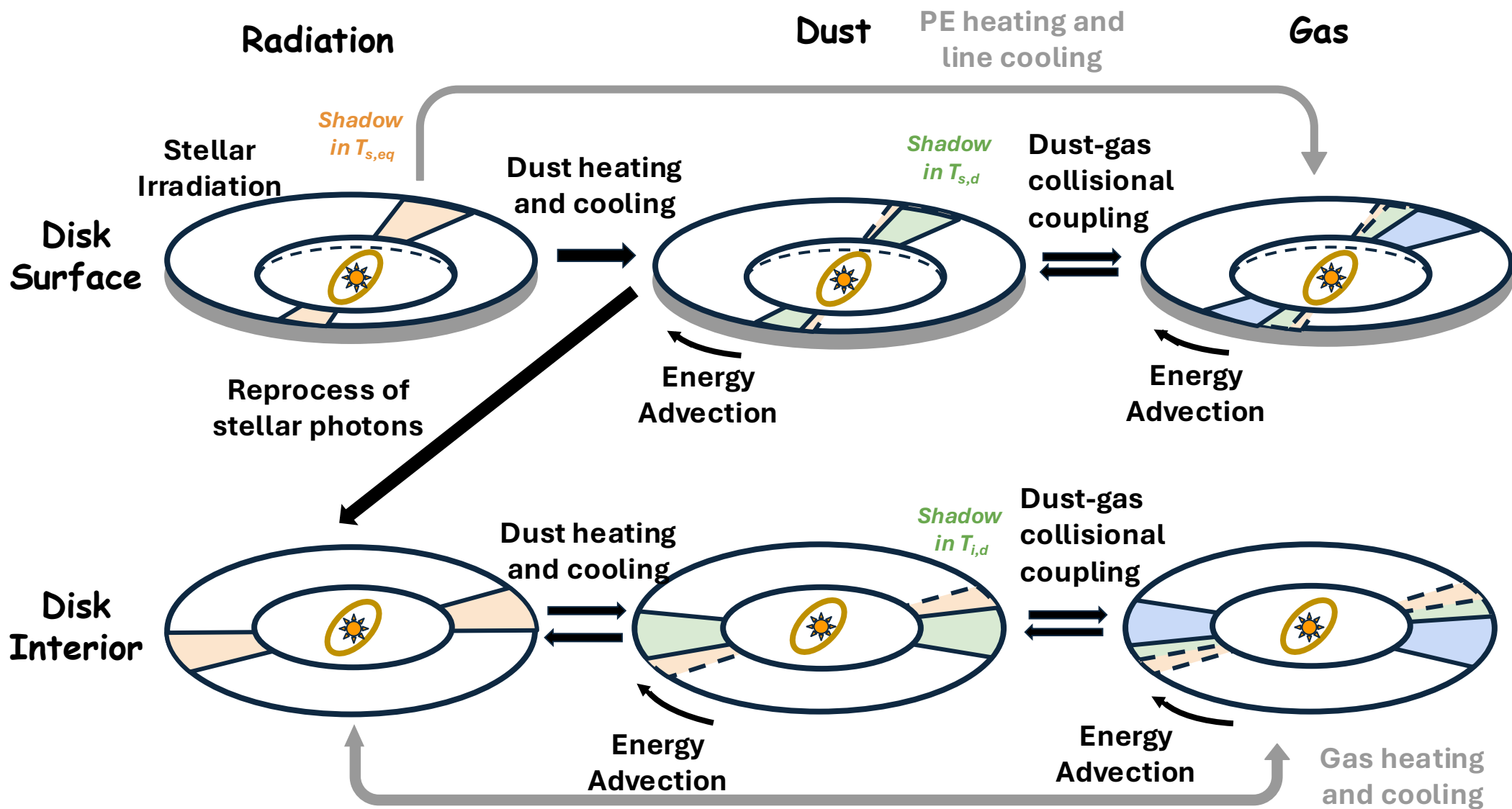


* Two armed spirals
come from the $m=2, n=0$
component in the full shadow
residual



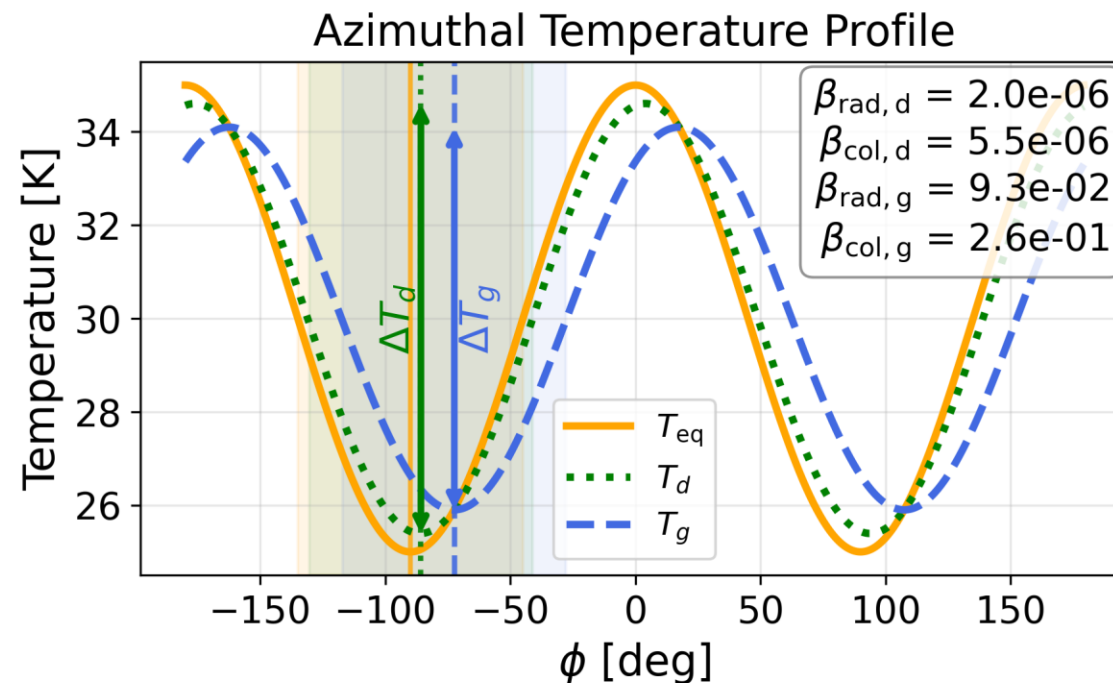
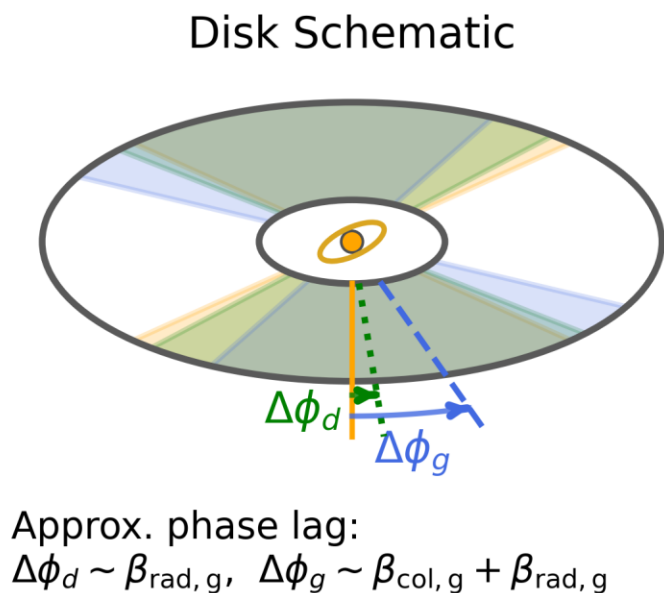
What about cooling time?





Shadows as probe of betas

Thermal Response to Shadow



$$\beta_{\text{rad},g}^{(\Sigma)} = \frac{\Sigma_g c_p \tau_{\text{eff}}}{8\sigma_{\text{SB}} T_{d,\text{eq}}^3} \Omega.$$

$$\beta_{\text{col},g}^{(\Sigma)} = \bar{\alpha}_T^{-1} \frac{2}{3} \frac{\gamma_g}{\gamma_g - 1} \frac{\Sigma_g}{\Sigma_d} \left[1 + \left(\frac{H_d}{H_g} \right)^2 \right]^{1/2} \text{St}_{\text{mid},S},$$

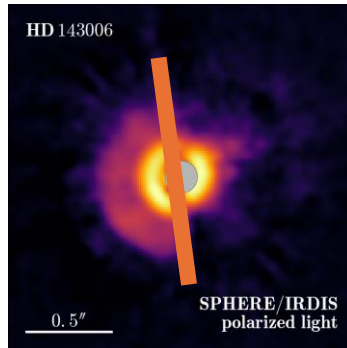
Joint fitting opportunities

Radiation

Dust

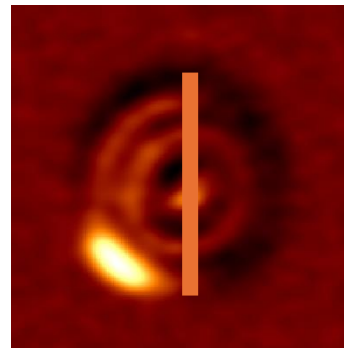
Gas

Disk
Surface

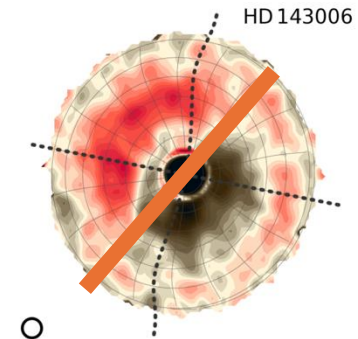


NIR scattered light
(Benisty+ 18)

Disk
Interior



Dust continuum
(exoALMA Curone+ 25)



^{12}CO temperature
(exoALMA
Fukagawa+ 26)

Summary

- **Disk's thermal structure and dynamics are tightly coupled**
- Misaligned shadows generate diverse substructures: spirals, gaps, rings, vortices, and can excite eccentricity and warps
- Misaligned shadows also drive strong accretion through spiral density waves and nozzle shocks, playing a critical role in disk evolution
- Current observations are well positioned to test these insights, enabling independent constraints on disk thermodynamics and dust properties